$$QD = \frac{1}{2}(Q_3 - Q_1)$$

$$= \frac{54.72 - 22.5}{2} = 16.11$$

$\therefore$ QD = 16.11

8.2.2 Mean deviation

Mean Deviation (MD) is defined as the average of the absolute difference between the items in a distribution and the mean or median of that series.

(i) Computation of Mean Deviation - Individual observations

If $X_1, X_2, X_3, ...X_n$ are n given observation then the mean deviation about mean or median is as follows

$$\text{MD about Mean} = \frac{\sum|X - \bar{X}|}{n} = \frac{\sum|D|}{n}$$

where $|D| = |X - \bar{X}|$ and n is the number of observations.

$$\text{MD about Median} = \frac{\Sigma|x - median|}{n} = \frac{\Sigma|D|}{n}$$

where $|D| = |X - median|$ and n is the number of observations.

NOTE

If the Mean deviation is computed from Median then in that case $|D|$ shall denote deviations of the items from Median, ignoring signs.

(ii) Computation of Mean Deviation - Discrete series

In discrete series the formula for calculating mean deviation is

$$\text{MD about Mean} = \frac{\sum f|X - \bar{X}|}{N} = \frac{\sum f|D|}{N}$$

where $|D| = |X - \bar{X}|$ by ignoring negative signs and N is total frequencies.

$$\text{MD about Median} = \frac{\sum f|X - Median|}{N}$$

$$= \frac{\sum f|D|}{N}$$

where $|D|$ = $|X$ – Median$|$ by ignoring negative sign and N is total frequencies.

(iii) Calculation of Mean Deviation- Continuous Series

For calculating Mean deviation in continuous series we have to obtain the midpoints of the various classes and take the deviations of these mid points from mean or median.

$$\text{MD about Mean} = \frac{\sum f|M - \bar{X}|}{N} \text{ (or)}$$

$$\text{MD about Mean} = \frac{\sum f|D|}{N}$$

where M is a mid value, $|D| = |M - \bar{X}|$ and N is the total frequencies.

$$\text{MD about Median} = \frac{\sum f|M - Median|}{N} \text{ (or)}$$

$$\text{MD about Mean} = \frac{\sum f|D|}{N}$$

where M is a mid value, $|D|$ = $|M$–Median$|$ (by ignoring negative sign) and N is the total frequencies.

(iv) Relative Measure for Mean Deviation

The relative measure corresponding to the mean deviation is called the coefficient of mean deviation and it is obtained as follows

$$\text{Coefficient of MD about mean} = \frac{Mean\ Deviation\ about\ Mean}{Mean}$$

$$\text{Coefficient of MD about median} = \frac{Mean\ Deviation\ about\ Median}{Median}$$

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